

*MA.7.AR.4.5***Benchmark****MA.7.AR.4.5 Solve real-world problems involving proportional relationships.**

Example: Gordy is taking a trip from Tallahassee, FL to Portland, Maine which is about 1,407 miles. On average his SUV gets 23.1 miles per gallon on the highway and his gas tanks holds 17.5 gallons. If Gordy starts with a full tank of gas, how many times will he be required to fill the gas tank?

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.AR.2.2
- MA.7.AR.3.1, MA.7.AR.3.2, MA.7.AR.3.3
- MA.7.GR.1.5
- MA.7.GR.2.2, MA.7.GR.2.3
- MA.7.DP.1.3, MA.7.DP.1.4
- MA.912.AR.2.3

Terms from the K-12 Glossary

- Proportional Relationships
- Rate
- Unit Rates

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.3.5

Next Benchmarks

- MA.8.AR.3.4, MA.8.AR.3.5

Purpose and Instructional Strategies

In grade 6, students solved mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system. In grade 7, students solve real-world problems involving proportional relationships. In grade 8, students will solve real-world problems involving linear relationships.

- This benchmark is a culmination of the work students have been doing throughout MA.7.AR.4.
- Instruction for this benchmark includes opportunities to compare two different proportional relationships to each other.
- Allow various methods for solving, encouraging discussion and analysis of efficient and effective solutions (*MTR.4.1*).
- In grade 7, it is not the expectation that students define independent and dependent variables. Students may have experience with these terms from other content areas.

Common Misconceptions or Errors

- Students may invert values when setting up proportions.

Strategies to Support Tiered Instruction

- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions.
 - What do you know from the problem?
 - What is the problem asking you to find?
 - What are the two quantities in this problem?
 - How are the quantities related to each other?
- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose.
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Instruction includes the understanding that the independent variable depends on the given context. Additionally, independent variables are not always the x -axis and the dependent variable are not always the y -axis.
 - For example, if one has a proportional relationship between feet and meters, they can graph feet either on the x -axis or the y -axis. Which one is dependent depends on the context. For instance, if one is given feet and converting to meters, then feet would be independent, and meters would be dependent.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

Patsy is making shortbread cookies using the ingredients below.

10 tablespoons of butter	$1\frac{1}{2}$ cups flour
$1\frac{1}{2}$ cups powdered sugar	$\frac{1}{2}$ teaspoon salt
$\frac{1}{2}$ teaspoon vanilla extract	

Part A. This recipe makes 16 cookies, but Patsy needs 5 dozen. How much of each ingredient will she need to make the 5 dozen cookies she needs?

Part B. Once Harrison tasted Patsy's shortbread cookies, he ordered 7 dozen for a birthday party. If Patsy originally started with 4 cups of flour, 2 cups of powdered sugar and 16 tablespoons of butter, how much more (if any) will she need of each ingredient to complete Harrison's order?

Part C. After the party, Jeb shared his recipe which calls for 2 cups of flour and $1\frac{3}{4}$ cup of powdered sugar. Since adding powdered sugar to cookies should make them sweeter, Jeb claims his larger ratio of powdered sugar to flour will produce sweeter cookies. Is this statement correct?

Instructional Items

Instructional Item 1

A couple is taking a horse and carriage ride through Central Park in New York City. After 8 minutes, they had traveled $\frac{1}{2}$ mile.

Part A. Create a graph to represent the proportional relationship between miles traveled and the number of minutes they are on the carriage.

Part B. Use this graph to determine how long it will take to complete the 2.5-mile ride around the park.

The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.